

Alper Kulturel

Financial Analyst · Data Analyst · Quantitative Analyst

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PROFESSIONAL SUMMARY

Financial Analyst with 2+ years of experience in financial data analysis, performance reporting, and KPI-driven decision support across finance, AI, and technology. In addition to analytical work, I develop the tools behind it, with dozens of public projects on GitHub covering automated reporting pipelines, risk models, and trading strategy backtests. Currently seeking Financial Analyst, Data Analyst, or Quantitative Analyst roles in the US.

CORE COMPETENCIES

Financial Analysis · Performance Reporting · Budget vs. Actuals & Variance Analysis · KPI Design & Forecasting · Credit Risk (PD/LGD/EAD) · Risk Modelling (VaR / CVaR) · Portfolio Optimisation · Strategy Backtesting · ETL & Data Pipelines · Interpretable ML

SELECTED PROJECTS

Quantitative Finance

options-pricing-engine — Option valuation by Black-Scholes, Monte Carlo and binomial lattice, with full Greeks, implied volatility and a volatility surface.

stock-backtesting-engine — Event-driven equity backtester modelling transaction costs and slippage, with equity, drawdown and rolling-Sharpe reporting.

pairs-trading — Cointegration-based market-neutral pairs strategy with ADF testing and a full backtest.

portfolio-optimization — Markowitz efficient frontier comparing Max Sharpe, Min Variance and Risk Parity against a naive 1/N benchmark.

monte-carlo-risk-simulator — 10,000 correlated future paths producing VaR, CVaR, maximum drawdown and risk-adjusted return.

sentiment-analyzer — Financial news sentiment (VADER, FinBERT) tested for predictive power against next-session returns.

market-daily — Automated end-of-day snapshot across equities, volatility, rates, FX and commodities.

Financial Analytics

credit-risk-scoring — PD/LGD/EAD framework producing Expected Loss, internal credit grades and Basel IRB capital.

fpa-variance-dashboard — Price/volume/mix attribution for budget versus actuals, with materiality scoring.

financial-reporting-pipeline — Validated ETL into PostgreSQL computing revenue, margin and EBITDA KPIs, emitting a four-sheet Excel report.

fpa-dashboard — Tableau FP&A dashboard over a multi-region budget-versus-actuals dataset.

market-data-etl — Twenty hand-written SQL queries over two years of S&P 500 daily data, portable across PostgreSQL and SQLite.

Data Analytics

banking-churn-predictor — Cross-validated churn model with class-imbalance handling and SHAP explanations.

sales-dashboard — RFM segmentation, cohort retention, ABC analysis and discount-impact modelling for retail.

crypto-data-pipeline — Live WebSocket trade ingestion with batching and deduplication into PostgreSQL, containerised.

Systems & Infrastructure

cpp-black-scholes — Dependency-free C++17 Black-Scholes library with pybind11 bindings; 38.8 ns per scalar price.

cpp-order-book — Price-time priority matching engine with $O(\log n)$ add and $O(1)$ cancel at ~67 ns per insert.

cpp-feed-handler — Low-latency ITCH binary feed parser with p50/p99 benchmarks and an optimisation ablation study.

EDUCATION

MSc, International Management & Information Systems

Fachhochschule Südwestfalen, Germany

BA, Business

Skidmore College, USA

- NCAA Division III All-American, Men's Tennis Doubles (2022)

CERTIFICATIONS

- Unity Certified Associate: Game Developer
- Microsoft Foundational C#

TECHNICAL SKILLS

Languages: Python, C++17, SQL, C#, JavaScript

Data & Analytics: pandas, NumPy, SciPy, scikit-learn, XGBoost, SHAP, statsmodels, Plotly, Streamlit, Tableau

Data Engineering: PostgreSQL, SQLAlchemy, Docker, WebSocket ingestion, ETL design, data validation

Systems: CMake, pybind11, low-latency profiling, binary protocol parsing